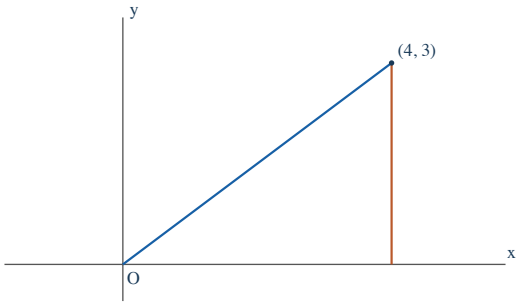
Polar Coordinates

What the devil are polar coordinates?

Solution 2

A polar point (r, θ) is a distance r from the pole, at angle θ anticlockwise from the initial line. Angles are in radians.

Convert the Cartesian point $(4, 3)$ to polar form.



Answer

$r = \sqrt{4^2 + 3^2} = 5, \quad \theta = \arctan(3/4) \approx 0.644$

The point is $(5, 0.644)$. Add multiples of 2π to obtain equivalent angles.

Converting to/from polar coordinates

Solution 3

Converting coordinates

$x = r\cos\theta,\ y = r\sin\theta,\ r^2 = x^2 + y^2$. Choose the angle from the correct quadrant.

Convert these Cartesian coordinates to polar form:
$(0, 2),\ (-3, 0),\ (1, 1),\ (-5, 12),\ (3\sqrt{3}, -3)$.

Polar coordinates

$(2, \pi/2),\quad (3, \pi),\quad (\sqrt{2}, \pi/4)$

$(13, \pi - \arctan(12/5)),\quad (6, -\pi/6)$

Polar to Cartesian equations

Solution 4

Convert to Cartesian form:

a. $r = 5$

b. $r = 2 + \cos 2\theta$

c. $r^2 = \sin 2\theta$

a

$x^2 + y^2 = 25$

b

$r^3 = 2r^2 + x^2 - y^2 = 3x^2 + y^2$

$(x^2 + y^2)^{3/2} = 3x^2 + y^2$

c

$r^4 = 2xy \Rightarrow (x^2 + y^2)^2 = 2xy$

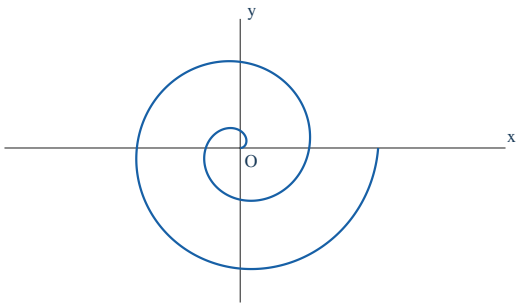
Why polar form sometimes are preferable

Solution 5

A spiral: $r = \theta, \theta \geq 0$.

As the angle increases, the distance from the pole increases.

Its parametric form is $x = \theta \cos\theta,\ y = \theta \sin\theta$.



Why keep polar form?

An inverse-trigonometric expression loses angle information: $\arccos(x/r)$ returns only angles from 0 to π. It does not describe the complete multi-turn spiral.

Cartesian to polar equations

Solution 6

Convert to polar form:

a. $y^2 = 4x$

b. $x^2 - y^2 = 5$

c. $\sqrt{3}y = x + 4$

a

$r\sin^2\theta = 4\cos\theta \Rightarrow r = 4\cot\theta\csc\theta$

The origin is included separately.

b

$r^2\cos 2\theta = 5 \Rightarrow r^2 = 5\sec 2\theta$

c

$r(\sqrt{3}\sin\theta - \cos\theta) = 4$

$r\sin(\theta - \pi/6) = 2$

Exercise 5A - Core Pure Year 2 Page 104 — Solution 7

2. Convert these polar coordinates to Cartesian coordinates:

a. $(6, \pi/6)$

b. $(6, -\pi/6)$

c. $(6, 3\pi/4)$

a

$(3\sqrt{3}, 3)$

b

$(3\sqrt{3}, -3)$

c

$(-3\sqrt{2}, 3\sqrt{2})$

Exercise 5A - Core Pure Year 2 Page 104 — Solution 8

3a. Convert $r = 2$

3b. Convert $r = 3\sec\theta$

3c. Convert $r = 5\csc\theta$

a

$x^2 + y^2 = 4$

b

$x = 3$

c

$y = 5$

3d. Convert $r = 4a\tan\theta\sec\theta$

3e. Convert $r = 2a\cos\theta$

3f. Convert $r = 3a\sin\theta$

d

$x^2 = 4ay$

e

$(x - a)^2 + y^2 = a^2$

f

$x^2 + (y - 3a/2)^2 = 9a^2/4$

Exercise 5A - Core Pure Year 2 Page 104 Solution 9

4a. Convert $x^2 + y^2 = 16$

a

$r = 4$

4b. Convert $xy = 4$

b

$r^2 = 8 \csc 2\theta$

4c. Convert $(x^2 + y^2)^2 = 2xy$

c

$r^2 = \sin 2\theta$

4d. Convert $x^2 + y^2 - 2x = 0$

d

$r = 2\cos\theta$

4e. Convert $(x + y)^2 = 4$

e

$r^2 = \dfrac{4}{1 + \sin 2\theta}$

4f. Convert $x - y = 3$

f

$r = \dfrac{3}{\sqrt{2}} \sec(\theta + \pi/4)$

Challenge 5A - Core Pure Year 2 Page 104 Solution 10

5. Prove the distance formula for polar points $P = (r_1, \theta_1)$, $Q = (r_2, \theta_2)$.

Answer

$PQ^2 = (r_2\cos\theta_2 - r_1\cos\theta_1)^2$

$PQ^2 = \ + (r_2\sin\theta_2 - r_1\sin\theta_1)^2$

$= r_1^2 + r_2^2 - 2r_1r_2\cos(\theta_1 - \theta_2)$

Take the non-negative square root. Alternatively, apply the cosine rule to triangle OPQ.

Sketching Curves of Polar Equations — Solution 11

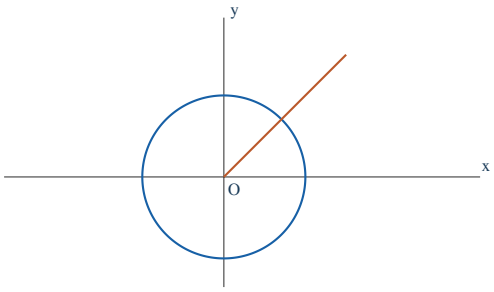
Recognising polar curves

$r = a$: circle, centre the pole, radius a.

$\theta = \alpha$, $r \geq 0$: a ray from the pole.

$r = a\theta$, $\theta \geq 0$: an Archimedean spiral.

Plot points and mark the direction of increasing angle.



Sketching using tables of values — Solution 12

Sketch $r = \sin 4\theta$, using $r \geq 0$.

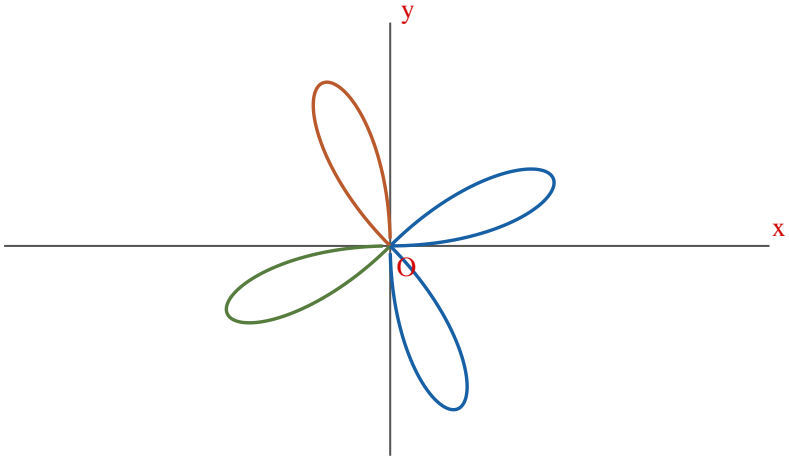
Start with $0 \leq \theta \leq \pi/4$.

Plot the loop

θ	0	$\pi/16$	$\pi/8$	$3\pi/16$	$\pi/4$
r	0	$1/\sqrt{2}$	1	$1/\sqrt{2}$	0

Repeat this positive loop after rotations of $\pi/2$. Negative values of r are excluded here.

Sketch



Sketching using tables of values

Solution 13

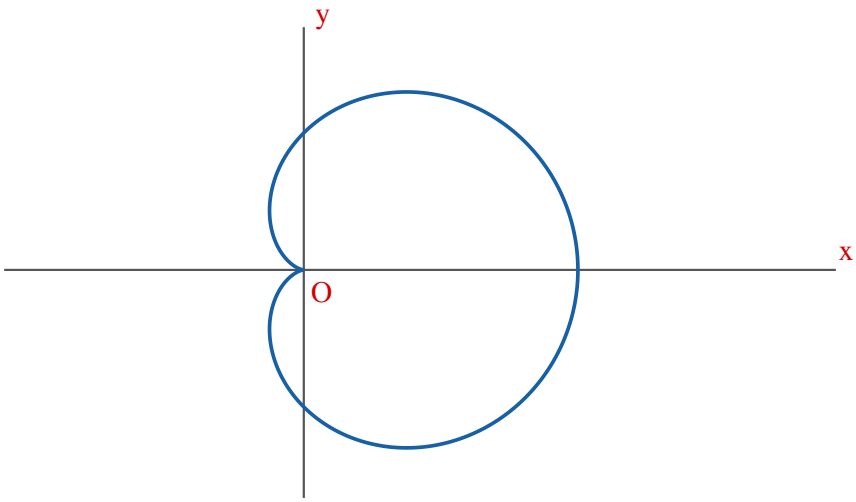
Sketch the cardioid $r = a(1 + \cos\theta)$, $a > 0$.

Key points

θ	0	$\pi/2$	π
r	$2a$	a	0

Reflect in the initial line. The pole is a cusp; the curve passes through $(2a, 0)$ and Cartesian points $(0, \pm a)$.

Sketch



More sketchies

Solution 14

Sketch:

a. $r = a\sin 3\theta$, $r \geq 0$.

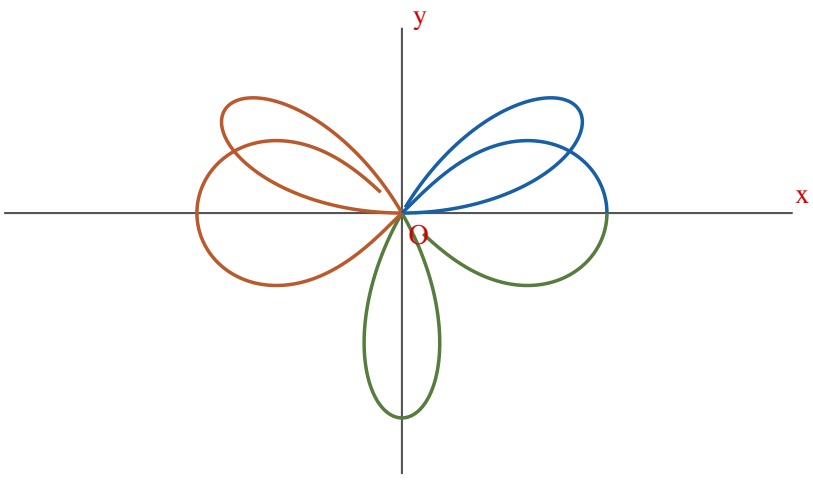
b. $r^2 = a^2\cos 2\theta$.

Answer

a. Three petals, with centres at angles $\pi/6, 5\pi/6, 3\pi/2$; maximum radius a.

b. A lemniscate: two loops about the initial line. Real radii require $\cos 2\theta \geq 0$; maximum radius a.

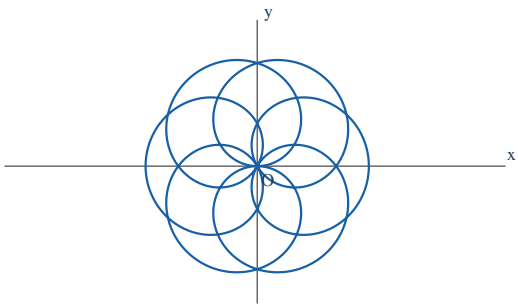
Sketch



Exploration

Plot $r = \cos(3\theta/4)$, allowing signed radii, for $0 \leq \theta \leq 8\pi$.

A negative radius is plotted in the opposite direction. This extension is separate from our usual $r \geq 0$ convention.



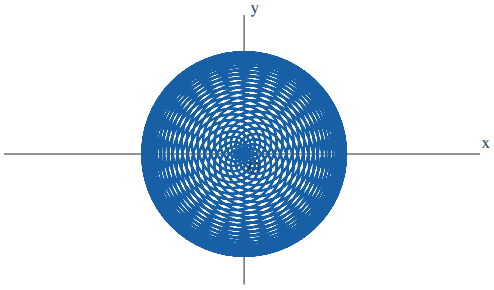
Observation

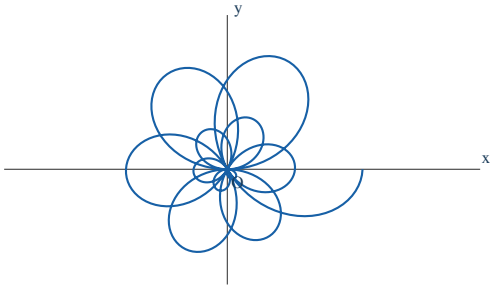
The trigonometric expression and the direction both repeat after 8π. Rational frequency ratios produce closed patterns.

More polar patterns

Compare $r = \cos(13\theta/99)$ with $r = \theta\cos(3\theta/2)$, allowing signed radii.

The first produces a closed multi-loop pattern; the second expands outward as the angle increases.



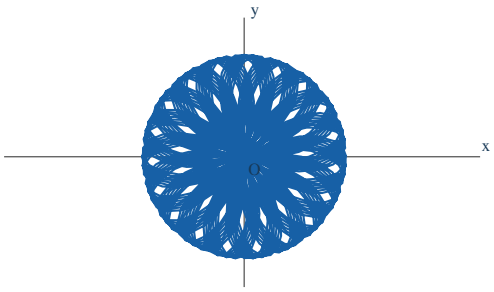


Just for fun... (not in the syllabus) — Solution 17

An irrational frequency

Plot $r = \cos(e\theta)$ for $0 \leq \theta \leq 50\pi$, allowing signed radii.

Since e is irrational, there is no common angular period that returns the full parametrisation to its starting state. The pattern does not close exactly.



Egg vs Dimple — Solution 18

Limaçons

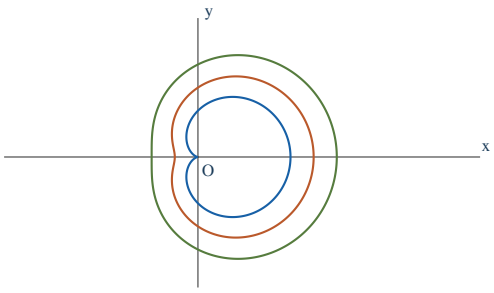
$r = a(p + q\cos\theta)$, with $a, q > 0, p \geq q$.

$p = q$: cardioid with a cusp.

$q < p < 2q$: dimpled limaçon.

$p \geq 2q$: convex limaçon.

The curve is symmetric in the initial line.



Egg vs Dimple

Solution 19

Sketch and compare:

a. $r = a(5 + 2\cos\theta)$

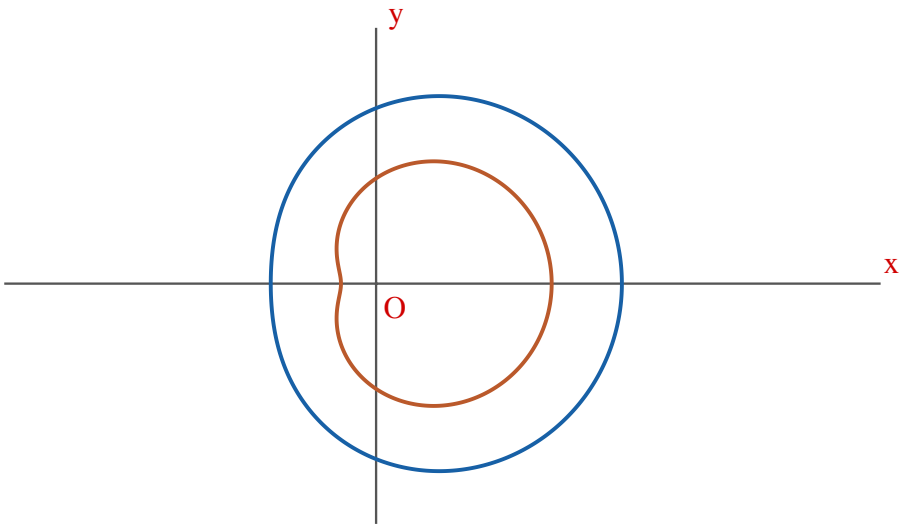
b. $r = a(3 + 2\cos\theta)$

Answer

a. Convex: radii at $0, \pi/2, \pi$ are $7a, 5a, 3a$.

b. Dimpled: corresponding radii are $5a, 3a, a$.

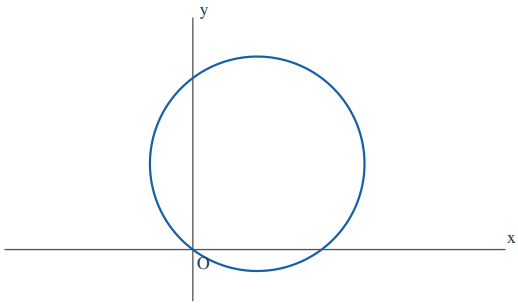
Sketch



Final Example

Solution 20

The locus $|z - 3 - 4i| = 5$ is a circle. Find its polar equation.



Answer

$(x-3)^2 + (y-4)^2 = 25$

$r^2 - 6r\cos\theta - 8r\sin\theta = 0$

$r = 6\cos\theta + 8\sin\theta$

The circle passes through the pole; include it when the radius is zero.

Exercise 5B - Core Pure Year 2 Pages 108-109 Solution 21

2. Sketch $r = k\sec(\pi/4 - \theta)$, $k > 0$, giving the axis intercepts.

Answer

$r(\cos\theta + \sin\theta)/\sqrt{2} = k$

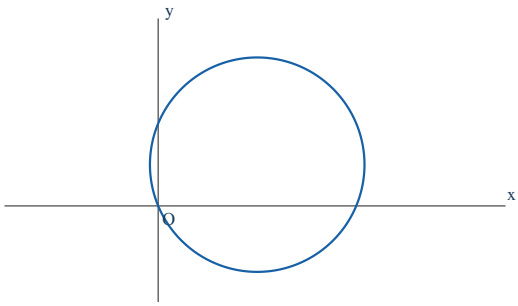
$x + y = \sqrt{2}k$

A straight line through $(\sqrt{2}k, 0)$ and $(0, \sqrt{2}k)$.

Exercise 5B - Core Pure Year 2 Pages 108-109 Solution 22

3a. Sketch $|z - 12 - 5i| = 13$ on an Argand diagram.

b. Show its polar equation is $r = 24\cos\theta + 10\sin\theta$.



a

A circle centred at $(12, 5)$, radius 13; it passes through the origin.

b

$(r\cos\theta - 12)^2 + (r\sin\theta - 5)^2 = 169$

$r^2 = 24r\cos\theta + 10r\sin\theta$

Divide by r away from the pole; the polar curve also includes the pole.

Curve summary

$r = a$: circle at the pole.

$\theta = \alpha$: ray at angle α.

$r = 2a\cos\theta$: circle centred at $(a, 0)$, radius a.

$r = k\theta$: spiral.

Always state the angular interval and whether signed radii are permitted.

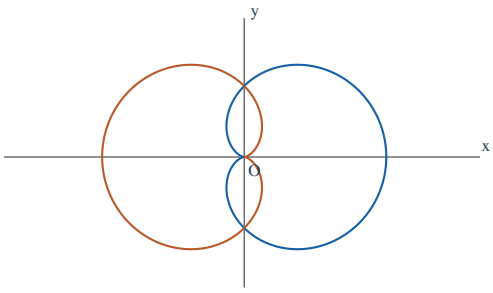
Symmetry and orientation

$a(1 + \cos\theta)$: cardioid facing right.

$a(1 - \cos\theta)$: cardioid facing left.

Replacing cosine by sine rotates the orientation.

$a(3 + 2\cos\theta)$: a dimpled limaçon, with minimum radius a.



More standard forms

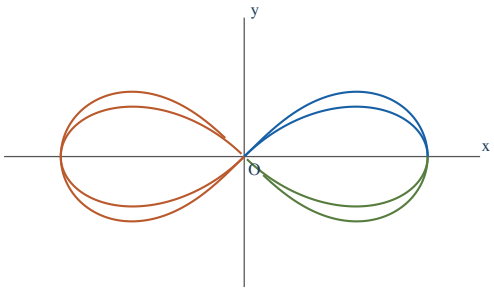
$r = a\cos 2\theta$, $r \geq 0$: two positive-radius petals.

$r^2 = a^2 \cos 2\theta$: lemniscate.

$r = p\sec(\alpha - \theta)$: straight line

$x\cos\alpha + y\sin\alpha = p$.

In particular, $r = a\sec\theta$ gives $x = a$.



From strips to sectors

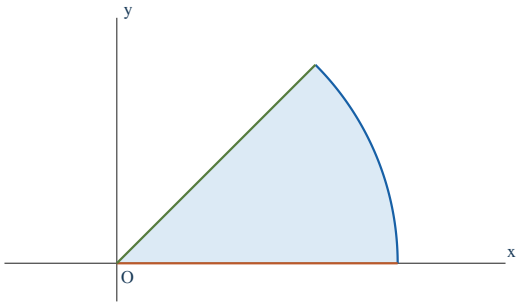
In Cartesian coordinates, area is accumulated from thin strips: $dA = y\,dx$.

A solid of revolution uses thin discs: $dV = \pi y^2 dx$.

For polar area, use thin sectors with angle $d\theta$.

Area in polar coordinates

A sector of radius r and angle $d\theta$ has area approximately $\frac{1}{2}r^2 d\theta$.

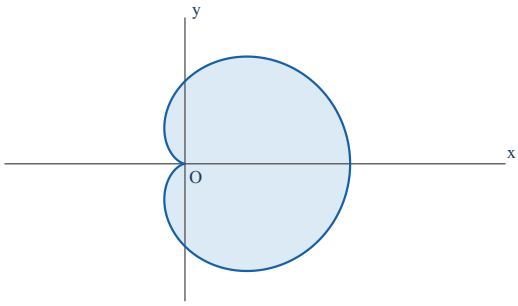


Formula

$$A = \frac{1}{2}\int_{\alpha}^{\beta} r^2\, d\theta$$

Choose limits that trace the required region once. Split the integral where the boundary changes; use symmetry when appropriate.

Find the area enclosed by $r = a(1+\cos\theta)$.



Answer

$$A = \frac{a^2}{2}\int_{-\pi}^{\pi}(1+2\cos\theta+\cos^2\theta)d\theta$$

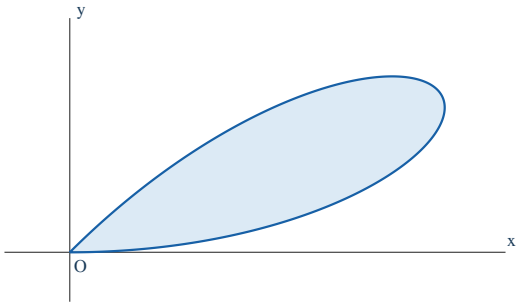
$$= \frac{a^2}{2}\left[\tfrac{3}{2}\theta + 2\sin\theta + \tfrac{1}{4}\sin 2\theta\right]_{-\pi}^{\pi}$$

$$= \frac{3\pi a^2}{2}$$

Another Example

Solution 29

Find the area of one loop of $r = a \sin 4\theta$, $r \geq 0$.



Answer

The first loop is traced for $0 \leq \theta \leq \pi/4$.

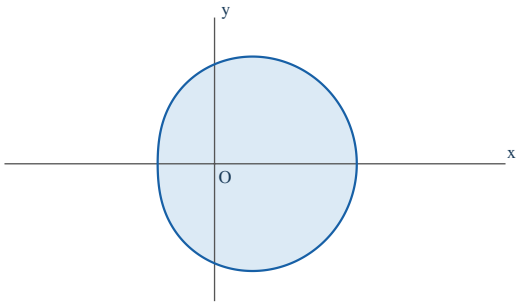
$$A = \frac{a^2}{2} \int_0^{\pi/4} \sin^2 4\theta \, d\theta$$

$$= \frac{a^2}{4} [\theta - \sin 8\theta/8]_0^{\pi/4} = \frac{\pi a^2}{16}$$

Test Your Understanding So Far

Solution 30

The area enclosed by $r = a + 3\cos\theta$, $a > 3$, is $107\pi/2$. Find a.



Answer

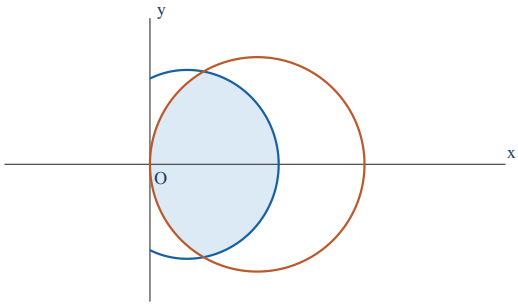
$$A = \frac{1}{2} \int_0^{2\pi} (a + 3\cos\theta)^2 d\theta = \pi a^2 + \frac{9\pi}{2}$$

$$2a^2 + 9 = 107 \Rightarrow a = 7$$

Intersecting Areas

Solution 31

Find the area common to $r = 2 + \cos\theta$ and $r = 5\cos\theta$.



Intersections

$2 + \cos\theta = 5\cos\theta \Rightarrow \theta = \pm\pi/3$

Area

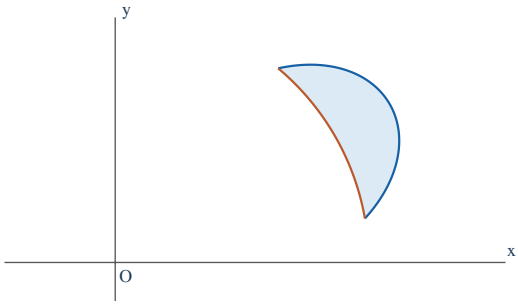
$$A = \int_0^{\pi/3} (2+\cos\theta)^2 d\theta + \int_{\pi/3}^{\pi/2} 25\cos^2\theta\, d\theta$$

$$= \frac{43\pi}{12} - \sqrt{3}$$

Test Your Understanding

Solution 32

Find the area outside $r = 2$ but inside $r = \frac{3}{2} + \sin 3\theta$, for $0 \le \theta \le \pi/2$.



Intersections $\sin 3\theta = \frac{1}{2} \Rightarrow \alpha = \pi/18, \quad \beta = 5\pi/18$

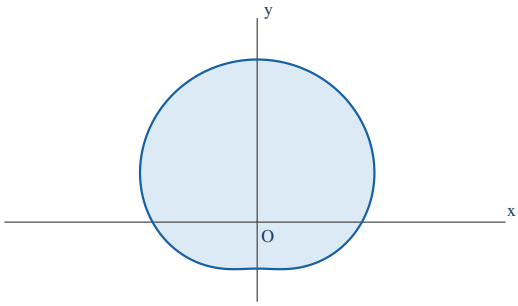
Area

$$A = \frac{1}{2}\int_\alpha^\beta [(\tfrac{3}{2} + \sin 3\theta)^2 - 4] d\theta$$

$$= \left[-\frac{5\theta}{8} - \frac{1}{2}\cos 3\theta - \frac{1}{24}\sin 6\theta\right]_\alpha^\beta$$

$$= \frac{13\sqrt{3}}{24} - \frac{5\pi}{36}$$

4. The area enclosed by $r = a + 5\sin\theta$, $a > 5$, is $187\pi/2$. Find a.

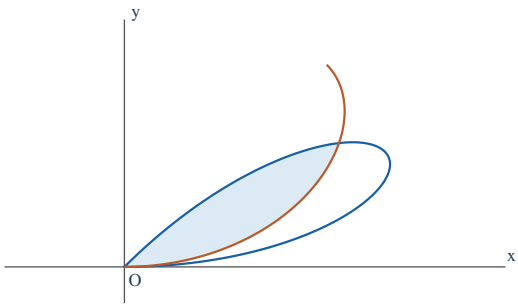


Answer

$A = \pi a^2 + 25\pi/2 = 187\pi/2$

$a^2 = 81 \Rightarrow a = 9$

5. Find the finite area common to $r = a\sin 4\theta$ and $r = a\sin 2\theta$, with $0 \le \theta \le \pi/2$ and $r \ge 0$.



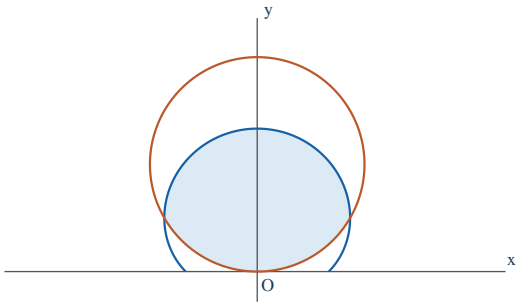
Intersections The positive loops meet at $\theta = \pi/6$, as well as at the pole.

Area

$$A = \frac{a^2}{2}\left(\int_0^{\pi/6} \sin^2 2\theta\, d\theta + \int_{\pi/6}^{\pi/4} \sin^2 4\theta\, d\theta\right)$$

$$= \frac{a^2}{4}\left(\frac{\pi}{4} - \frac{3\sqrt{3}}{16}\right)$$

6. Find the area common to $r = 1 + \sin\theta$ and $r = 3\sin\theta$.



Intersections

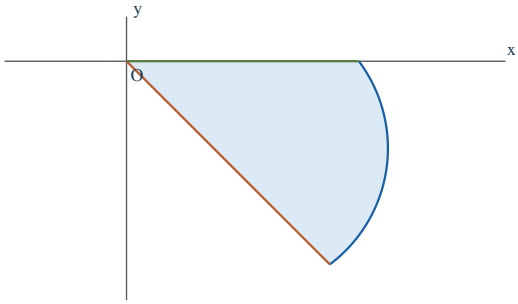
$\sin\theta = \frac{1}{2} \Rightarrow \theta = \pi/6, 5\pi/6$

Use symmetry

$$A = \int_0^{\pi/6} 9\sin^2\theta\, d\theta + \int_{\pi/6}^{\pi/2} (1+\sin\theta)^2 d\theta$$

$$= \frac{5\pi}{4}$$

7. Sketch the region $-\pi/4 \le \arg z \le 0$, $|z - 4 + 3i| \le 5$. Find its area to 3 significant figures. The circle has polar equation $r = 8\cos\theta - 6\sin\theta$.



Sketch

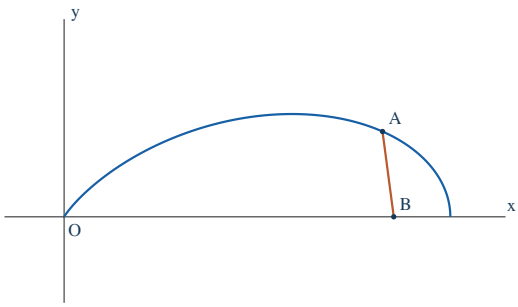
Take the part of the disc centred at $(4, -3)$, radius 5, between the rays $\theta = -\pi/4$ and 0.

Area

$$A = \frac{1}{2}\int_{-\pi/4}^{0} (8\cos\theta - 6\sin\theta)^2 d\theta$$

$$= \frac{31}{2} + \frac{25\pi}{4} \approx 35.1$$

9. On $r = 1 + \cos 3\theta$, $0 \le \theta \le \pi/3$, point A has radius $(2+\sqrt{2})/2$. Point B is on the initial line with the same radius. Find the area between arc A to $(2, 0)$, AB and the initial line, to 3 significant figures.



Sector

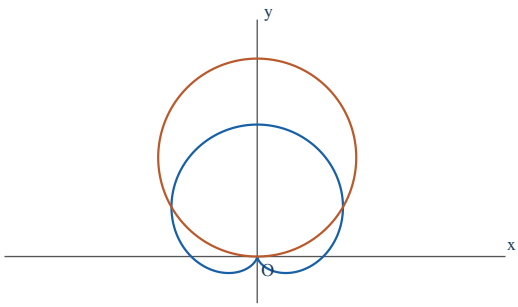
$\theta_A = \pi/12, \quad A_{\text{sector}} = \pi/16 + \sqrt{2}/6 + 1/24$

Subtract triangle

$$A_{\triangle OAB} = \frac{1}{2}\left(\frac{2+\sqrt{2}}{2}\right)^2 \sin(\pi/12)$$

$A = A_{\text{sector}} - A_{\triangle OAB} \approx 0.0966$

10. Find the area inside the cardioid $r = 1 + \sin\theta$ and outside the circle $r = 3\sin\theta$, to 2 decimal places.



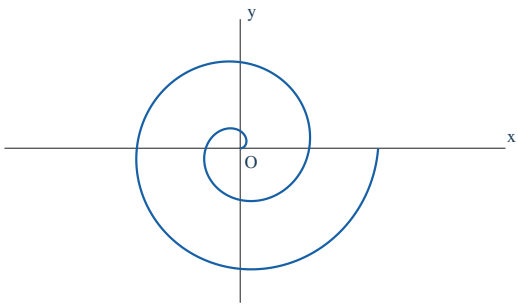
Answer

The cardioid has total area $3\pi/2$. The common area, as in question 6, is $5\pi/4$.

$A = 3\pi/2 - 5\pi/4 = \pi/4 \approx 0.79$

Challenge 5C - Core Pure Year 2 Pages 111-112 Solution 39

Challenge. A shell is modelled by $r = k\theta$, $0 \le \theta \le 4\pi$. The width along the initial line, from the point at 3π to the point at 4π, is 3 cm. Find k and the area inside the outer turn.



Scale

$$3\pi k + 4\pi k = 3 \Rightarrow k = \frac{3}{7\pi}$$

Area

Use only the outer turn so the inner region is not counted twice.

$$A = \frac{1}{2}\int_{2\pi}^{4\pi} \left(\frac{3\theta}{7\pi}\right)^2 d\theta = \frac{12\pi}{7} \text{ cm}^2$$

Tangents and Normals Solution 40

Tangents to polar curves

$x = r\cos\theta, \qquad y = r\sin\theta$

Differentiate

$$\frac{dy}{dx} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$$

Horizontal: numerator zero and denominator nonzero.

Vertical: denominator zero and numerator nonzero.

If both vanish, inspect the limiting tangent separately.

Example

Solution 41

Find the regular points where $r = a(1 + \cos\theta)$ has horizontal tangents.

Answer

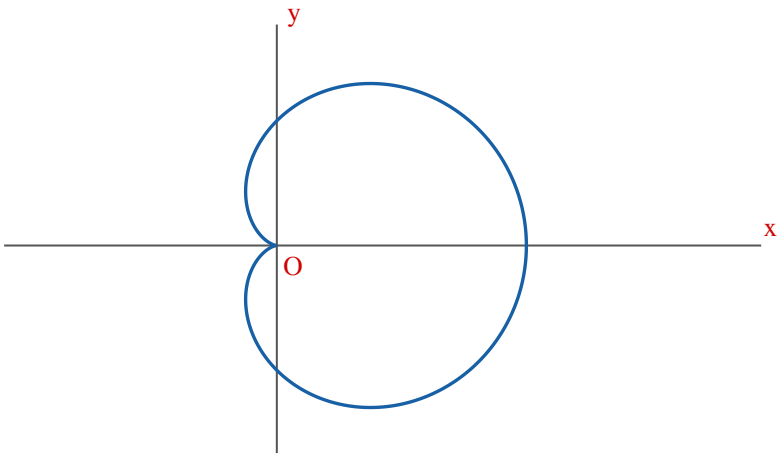
$$\frac{dy}{d\theta} = a(\cos\theta + \cos 2\theta)$$

$$(2\cos\theta - 1)(\cos\theta + 1) = 0$$

$$(r, \theta) = (3a/2, \pm\pi/3)$$

The pole at $\theta = \pi$ is a cusp, with a horizontal limiting tangent.

Sketch



Test Your Understanding So Far

Solution 42

For $r = 1 + 2\cos\theta$, $0 \le \theta \le \pi/2$, find the radius at the point with a horizontal tangent.

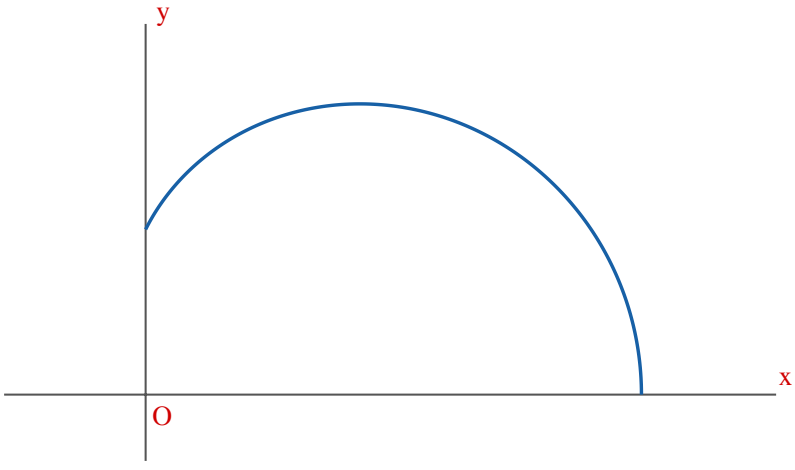
Answer

$$\frac{dy}{d\theta} = \cos\theta + 2\cos 2\theta = 0$$

$$4\cos^2\theta + \cos\theta - 2 = 0$$

$$\cos\theta = \frac{-1 + \sqrt{33}}{8}, \quad r = \frac{3 + \sqrt{33}}{4}$$

Sketch



Example

Solution 43

Find horizontal tangents to $r = a\sin 2\theta$, $0 \le \theta \le \pi/2$.

Answer

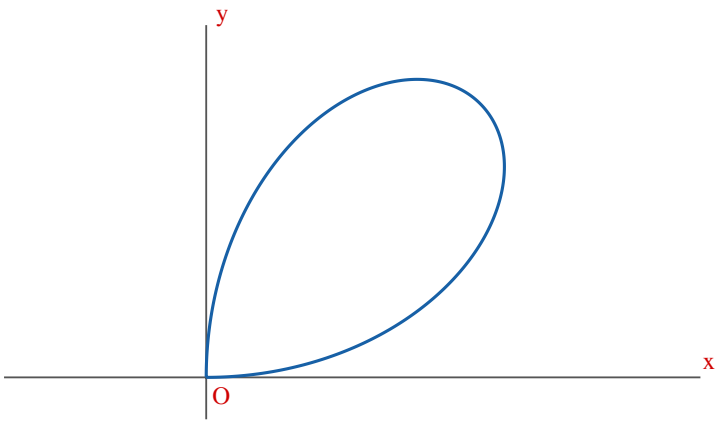
$y = 2a\sin^2\theta\cos\theta$

$\dfrac{dy}{d\theta} = 2a\sin\theta(2 - 3\sin^2\theta)$

At $\theta = 0$ the tangent is $y = 0$. The other point has $\theta = \arctan\sqrt{2}$, $r = 2a\sqrt{2}/3$.

$y = \dfrac{4a}{3\sqrt{3}}$

Sketch



Example

Solution 44

Find vertical tangents to $r = a\sin 2\theta$, $0 \le \theta \le \pi/2$.

Answer

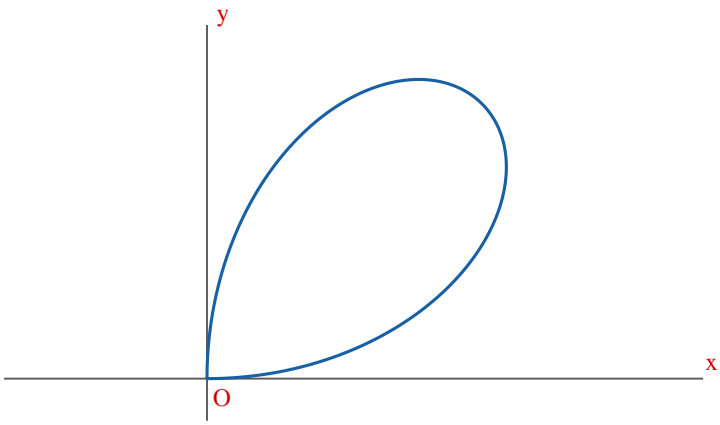
$x = 2a\sin\theta\cos^2\theta$

$\dfrac{dx}{d\theta} = 2a\cos\theta(1 - 3\sin^2\theta)$

At $\theta = \pi/2$, the tangent is $x = 0$. Otherwise $\theta = \arctan(1/\sqrt{2})$, $r = 2a\sqrt{2}/3$.

$x = \dfrac{4a}{3\sqrt{3}}$

Sketch



Proof of dimple vs egg

Solution 45

Explain why $r = a(p + q\cos\theta)$, $p > q > 0$, has a dimple when $p < 2q$.

Answer

$x = a(p\cos\theta + q\cos^2\theta)$

$\dfrac{dx}{d\theta} = -a\sin\theta(p + 2q\cos\theta)$

Besides $\theta = 0, \pi$, an extra pair of vertical tangents occurs when $\cos\theta = -p/(2q)$.

This is possible precisely when $q < p < 2q$, producing the dimple.

Exercise 5D - Core Pure Year 2 Pages 115-116

Solution 46

4. Find the polar coordinates on $r = a(7 + 2\cos\theta)$ where the tangents are parallel to the initial line.

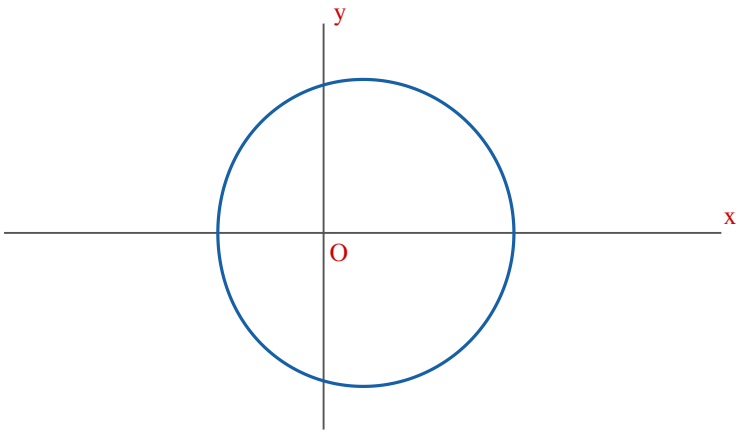
Answer

$\dfrac{dy}{d\theta} = a(7\cos\theta + 2\cos 2\theta) = 0$

$(4\cos\theta - 1)(\cos\theta + 2) = 0$

$(r, \theta) = \left(\dfrac{15a}{2}, \pm\arccos\dfrac{1}{4}\right)$

Sketch



5. Find the tangent lines to $r = 2 + \cos\theta$ that are perpendicular to the initial line.

Answer

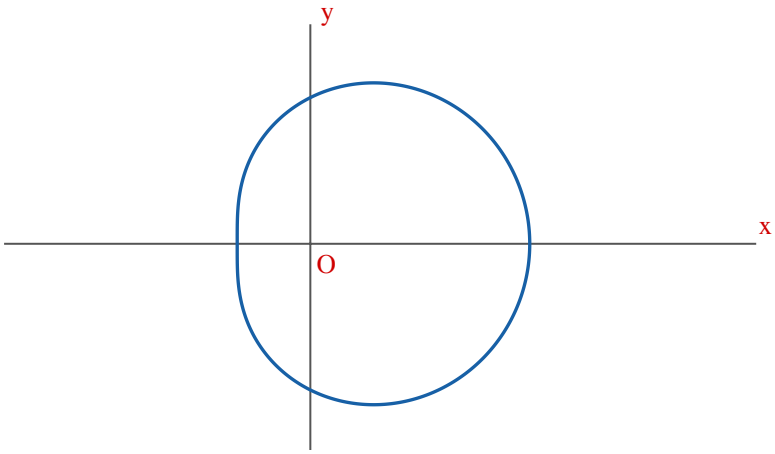
$\frac{dx}{d\theta} = -2\sin\theta(1 + \cos\theta) = 0$

At $\theta = 0, \pi$, the Cartesian coordinates have $x = 3, -1$.

$x = 3, \quad x = -1$

Equivalently, $r\cos\theta = 3$ or $r\cos\theta = -1$.

Sketch



6. Find the point on $r = a(1 + \tan\theta)$, $0 \le \theta < \pi/2$, with a tangent perpendicular to the initial line.

Answer

$x = a(\cos\theta + \sin\theta)$

$\frac{dx}{d\theta} = a(\cos\theta - \sin\theta) = 0$

$\theta = \pi/4, \quad (r, \theta) = (2a, \pi/4)$

7. A point A on $r = 1 + 3\cos\theta$, $0 \leq \theta \leq \pi/2$, has a horizontal tangent. Find the exact length OA.

Answer

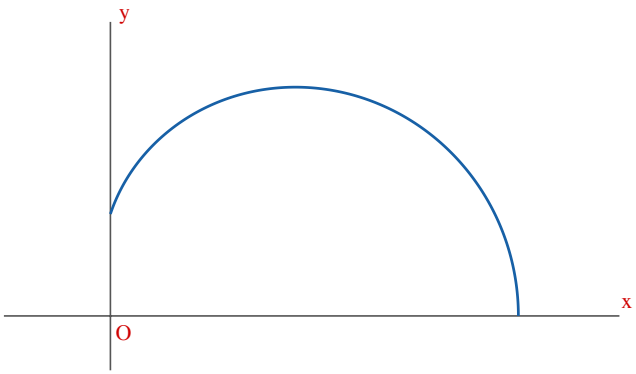
$$\frac{dy}{d\theta} = \cos\theta + 3\cos 2\theta = 0$$

$$6\cos^2\theta + \cos\theta - 3 = 0$$

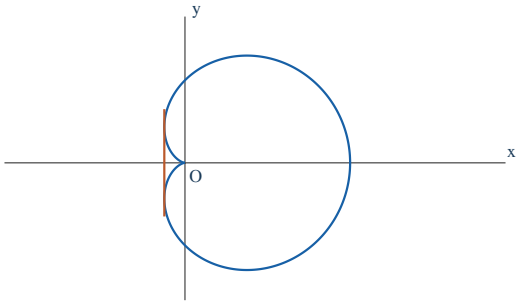
$$\cos\theta = \frac{-1 + \sqrt{73}}{12}$$

$$OA = r = \frac{3 + \sqrt{73}}{4}$$

Sketch



8. For the cardioid $r = 2(1 + \cos\theta)$, find the area between the curve and its left-hand vertical common tangent, to 3 significant figures.



Tangencies

$$\frac{dx}{d\theta} = -2\sin\theta(1 + 2\cos\theta) = 0$$

The required tangencies are $\theta = \pm 2\pi/3$, with $x = -1/2$.

Area

$$A = 2\left(\frac{\sqrt{3}}{8} - \frac{1}{2}\int_{2\pi/3}^{\pi}[2(1+\cos\theta)]^2 d\theta\right)$$

$$= \frac{15\sqrt{3}}{4} - 2\pi \approx 0.212$$